

ANVESHA

AI-assisted Kidney Stone Intelligence & Personal Health
Monitoring Platform

USER MANUAL &
FEATURE GUIDE

1 What is ANVESHA?

ANVESHA is a personal health monitoring and intelligent medical scan platform designed to help patients, family caregivers, and physicians easily track, analyze, and manage kidney health records and Non-Contrast Computed Tomography (NCCT) abdomen CT scans.

Kidney stones (calculi) are a common health concern. ANVESHA makes complex medical scan information easy to understand by providing simple visual summaries, calculus metrics (size, density, location), hydronephrosis grading, and printable doctor report summaries.

2 Key Benefits

For Patients & Families

- **Easy Understanding:** No medical degree required. Translates raw CT scans into simple numbers and clear guides.
- **Family Health Profiles:** Keep track of multiple family members' health records in one safe place.
- **Printable Doctor Reports:** Download or print ready-to-use medical summary reports for physician visits.
- **Safety Alerts:** Instant warnings to help identify when emergency medical care is needed.

For Healthcare Providers

- **Automated Scan Intake:** Fast processing of DICOM CT scan volumes (.dcm, .zip, .nii).
- **Calculus Metrics:** Calculates maximum stone size (mm), Hounsfield Unit (HU) density, and hydronephrosis grade.
- **Axial Slice Viewer:** Interactive slice simulator with Bone/Stone and Soft Tissue contrast presets.
- **Digital Review Sign-off:** Save clinical impression notes with secure digital verification signatures.

3 How ANVESHHA Works: Step-by-Step Guide

Using ANVESHHA is straightforward. Follow these steps to navigate and use all platform features:

Step 1: Personal Health Overview (Home / Dashboard)

DASHBOARD

When you log in to ANVESHHA, you start on the **Home Dashboard**. Here you can view:

- **Health Profiles:** Total family and personal profiles registered.
- **CT Scans Analyzed:** Number of completed scan evaluations.
- **Scans Awaiting Review:** Scans currently being reviewed by a doctor.
- **High-Risk Calculi Warning Cards:** Highlights stones larger than 7.0mm or scans flagged with significant kidney blockage (hydronephrosis).

Step 2: Managing Health Profiles (My Health Records)

PATIENTS TAB

Click on **My Health Records** in the side menu to view and add profile records:

- Click **" + Add New Health Profile "** to register a family member or personal profile.
- Enter profile name, age, gender, medical record number (MRN), contact number, and medical history.
- Filter profile records by risk category (*Critical, High, Moderate, Low*).

Step 3: Scan Upload & Intake (Scan Upload)

UPLOAD TAB

To analyze a new CT scan file:

- Click **Scan Upload** on the sidebar.
- Drag & drop your CT scan file (accepted formats: .dcm, .zip, .nii, max 500 MB).
- Click **"Start Automated Scan Intake"**. ANVESHHA will validate DICOM metadata headers, spatial resolution, and slice thickness.
- Alternatively, click **"Load Pre-Configured DICOM Sample"** to test the features instantly.

Step 4: Interactive Slice Visualizer & Findings (CT Analysis)

FINDINGS TAB

Explore your CT scan interactively:

- **Slice Slider:** Scroll slice-by-slice through the axial CT cross-section to view kidney anatomical structures.
- **Contrast Presets:** Toggle between *Bone/Stone (W:1500 L:400)* to highlight hard stones and *Soft Tissue (W:400 L:40)* to inspect kidney tissue.
- **ROI Bounding Box:** Highlight the exact location of detected kidney stones.
- **Calculus Metrics Card:** Review key measurements including Stone Location (e.g. *Right Lower Pole Calyx*), Maximum Size (mm), Density (HU), and Hydronephrosis Grade.

Step 5: Summary Report & Doctor Export (Reports)

REPORTS TAB

Access the complete printable report:

- Click "**Print Official Summary Report**" to generate a clean paper copy or PDF document for your doctor's appointment.
- Includes clinical impression notes, recommended treatments (e.g., fluid protocols, tamsulosin, urology consultation), and digital sign-off verification.

4 Simple Medical Glossary for Patients

Here is a quick reference table explaining common medical terms used in CT scan reports:

Medical Term	What It Means in Plain Language
NCCT Abdomen & Pelvis	Non-Contrast Computed Tomography scan. A 3D X-ray scan of the belly taken without injecting dye. It is the gold-standard scan for detecting kidney stones.
Calculus / Calculi	The medical term for a kidney stone (or multiple stones).
Hounsfield Units (HU)	A measurement of density. Soft tissues are around 30-50 HU, while hard calcium stones are usually 600 to 1200+ HU. Higher density stones are harder to break.
Hydronephrosis	Swelling of the kidney caused when a stone blocks the flow of urine down the ureter tube. Graded from Grade 1 (Mild) to Grade 4 (Severe).
Ureteric Dilatation	Wideness or swelling of the urine tube connecting the kidney to the bladder due to a downstream stone obstruction.
Medical Expulsive Therapy (MET)	Medications (such as Tamsulosin) prescribed by doctors to relax muscles in the ureter and help small stones (< 5-6mm) pass naturally in urine.

URGENT MEDICAL SAFETY NOTICE

ANVESHHA provides health tracking and educational insights. It is not a replacement for professional medical advice or emergency care.

Seek Emergency Hospital Care Immediately if you experience:

- Severe side or back (flank) pain accompanied by high fever or chills
- Severe nausea, vomiting, or total inability to pass urine
- Visible blood in urine combined with severe fever